FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE (Main)-2016

FULL TEST – II

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

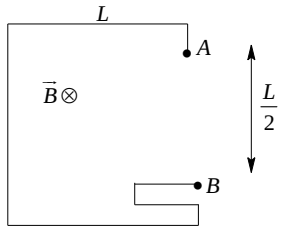
PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A bent wire AB carrying a current I is placed in a region of uniform magnetic field $\vec{B}$. The force on the wire AB is



(A) BIL

(B) $\frac{3}{2}BIL$

(C) Zero

(D) $\frac{BIL}{2}$

2. A conductor AB carries a current i in a magnetic field $\vec{B}$. If $\overrightarrow{AB}=\vec{r}$ and the force on the conductor is $\vec{F}$. Then:

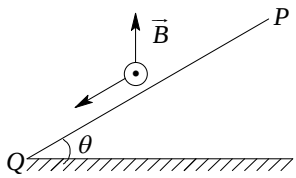
(A) $\vec{F}=\left(\vec{r}\times\vec{B}\right)i$

(B) $\vec{F}=i\left(\vec{B}\times\vec{r}\right)$

(C) $\left|\vec{F}\right|=i\left(\vec{r}\cdot\vec{B}\right)$

(D) $\vec{F}$ is the resultant effect of the electric forces on the electrons in the conductor.

3. A copper rod of length L and mass m is sliding down a smooth inclined plane of inclination θ with a constant speed υ. A current I is flowing in perpendicular to the plane of diagram inwards. A vertically upward magnetic field $\vec{B}$ exists in this region. The magnitude of the required magnetic field $\vec{B}$ is



(A) $\frac{mg}{IL}\sin\theta$

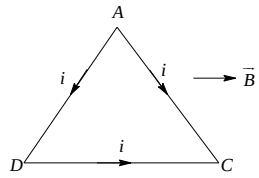
(B) $\frac{mg}{IL}\cos\theta$

(C) $\frac{mg}{IL}\tan\theta$

(D) $\frac{mg}{IL\sin\theta}$

Space for Rough work

4. An equilateral triangular loop ADC of side l carries a current i in the direction shown in figure. The loop is kept in a uniform horizontal magnetic field $\vec{B}$ as shown in figure. Net force on the loop is



(A) Zero

(B) $\frac{2}{3}ilB$, parallel to the plane of the loop

(C) $\frac{\sqrt{2}}{3}ilB$, perpendicular to the plane of loop, inwards

(D) $\sqrt{3}\,ilB$, perpendicular to the plane of loop

5. An iron wire of length l is magnetized have a magnetic moment m. If it is bent into a semicircle of radius r, its magnetic moment is.

(A) m (B) $\frac{2m}{\pi}$

(C) $\frac{m}{\pi}$ (D) $m\frac{1}{r}$

6. A magnet is suspended horizontally in the earth's magnetic field. When slightly displaced, it oscillates in the horizontal plane with a period T. If a piece of wood having the same moment of inertia as the magnet is attached to the magnet, the new period of oscillation is

(A) $\frac{T}{\sqrt{2}}$ (B) $\frac{T}{2}$

(C) $\sqrt{2}\,T$ (D) $2T$

7. An air core solenoid has 1000 turns and is one metre long. Its cross- sectional area is $10\,cm^2$. its self- inductance is

(A) $2.56\,mH$ (B) $1.92\,mH$

(C) $1.26\,mH$ (D) $2.78\,mH$

Space for Rough work

8. A capacitor of $10\,\mu F$ and an inductor of $1H$ are joined in series. An ac of $50\,Hz$ is applied to this combination. The impedance of the combination is

(A) $5.67\,\Omega$ (B) $4.16\,\Omega$

(C) $7.02\,\Omega$ (D) $3.21\,\Omega$

9. A light ray is incident an angle $45°$ with the normal to a $\sqrt{2}\,cm$ thick transparent plate $(n=2)$. The lateral shift of the light as it emerges out from the plate is

[Given: Lateral shift, $S_L = \dfrac{t\sin(i-r)}{\cos r}$]

(A) $\dfrac{\left(\sqrt{7}-1\right)}{\sqrt{7}}cm$ (B) $\left(\sqrt{7}-1\right)cm$

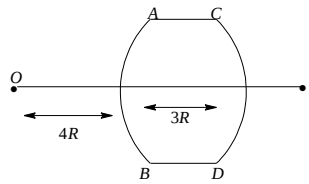
(C) $\dfrac{\left(\sqrt{7}-1\right)}{\left(\sqrt{7}+1\right)}cm$ (D) $\dfrac{\sqrt{2}-1}{\sqrt{2}}cm$

10. A fish is rising up vertically inside a pond with a velocity $4\,cm\,s^{-1}$, notices a bird, which is diving downwards. The velocity of the bird appears to be $16\,cm\,s^{-1}$ to the fish. The actual velocity of the diving bird is(refractive index of water is $\dfrac{4}{3}$)

(A) $12\,cm/s$ (B) $9\,cm/s$

(C) $8\,cm/s$ (D) $7.5\,cm/s$

11. Two surfaces AB *and* CD of a thick lens have radii of curvature R *and* $2R$ respectively. Thickness at the centre is $3R$. Refractive index of the material of the lens is 1.5. The image of a point object placed on the axis of the lens at a distance $4R$ from the surface of smaller radius of curvature and to its left is formed at

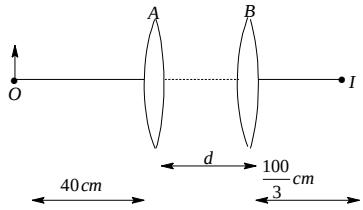


(A) $\dfrac{4R}{3}$ from CD to its right (B) $\dfrac{4R}{3}$ from CD to its left

(C) $\dfrac{8R}{3}$ from CD to its right (D) $\dfrac{8R}{3}$ from CD to its left

Space for Rough work

12. Two convergent lenses A *and* B (erect) of focal lengths $30\,cm$ *and* $50\,cm$ are separated by $d\,cm$ apart co- axially. An erect object of height $2\,cm$ is kept in front of I lens at $40\,cm$ from it. The real image is formed at $\frac{100}{3}\,cm$ to the right of II lens (see figure). The value of d is



(A) $20\,cm$ (B) $15\,cm$

(C) $10\,cm$ (D) $25\,cm$

13. Radius of gyration of a body about an axis at a distance $6\,cm$ from its centre of mass is $10\,cm$. its radius of gyration about a parallel axis through its centre of mass is

(A) $6\,cm$ (B) $8\,cm$

(C) $10\,cm$ (D) $12\,cm$

14. Moment of inertia of uniform rod of mass m and length l is $\frac{17}{12}ml^2$ about a line perpendicular to the rod. The distance of this line from the middle point of the rod is

(A) l (B) $\frac{l}{2\sqrt{3}}$

(C) $\frac{l}{\sqrt{2}}$ (D) $\frac{2}{3}l$

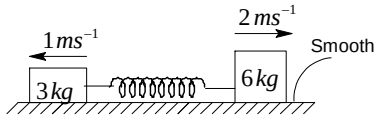
15. Two particles of masses m *and* $2m$ moving in opposite directions collide elastically with velocities υ *and* 2υ respectively. The velocity of $2\,m$ after collision is:

(A) υ (B) 1.5υ

(C) Zero (D) 1.1υ

Space for Rough work

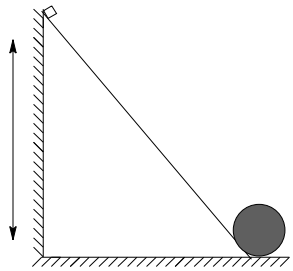
16. Two blocks of masses $3\,kg$ *and* $6\,kg$ are placed on a smooth horizontal surface. They are connected by a light spring of force constant, $k = 200\,Nm^{-1}$. Initially the spring is unstretched. The indicated velocities are imparted to the blocks. The maximum extension of the spring is



(A) $10\,cm$ (B) $15\,cm$

(C) $20\,cm$ (D) $30\,cm$

17. A block is released from rest at the top of an inclined plane which later curves into a circular track of radius r as shown in figure. Find the minimum height h form where it should be released so that it is able to complete the circle.



(A) $2.5\,r$ (B) $3\,r$

(C) $2\,r$ (D) $3.5\,r$

18. A body of mass m, having momentum p, is moving on a rough horizontal surface. If it is stopped in a distance x, the coefficient of friction between the body and the surface is given by:

(A) $\mu = \dfrac{p^2}{2\,gm^2x}$ (B) $\mu = \dfrac{p^2}{2\,gmx}$

(C) $\mu = \dfrac{p}{2\,gmx}$ (D) $\mu = \dfrac{p}{2\,gm^2x}$

19. Two equal masses are attached to the two ends of a spring of spring constant k. The masses are pulled apart symmetrically to stretch the spring by a length x over its natural length. The work done by the spring on each mass is:

(A) $\dfrac{1}{2}kx^2$ (B) $-\dfrac{1}{2}kx^2$

(C) $\dfrac{1}{4}kx^2$ (D) $-\dfrac{1}{4}kx^2$

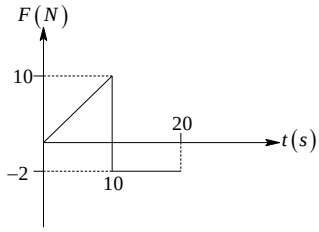
Space for Rough work

20. A net force of $200\ N$ gives a body of mass m_1 an acceleration of $80\ ms^{-2}$ and a body of mass m_2, an acceleration of $240\ ms^{-2}$. The acceleration that this force caused when the masses combine together is

(A) $50\ ms^{-2}$ (B) $60\ ms^{-2}$

(C) $120\ ms^{-2}$ (D) $100\ ms^{-2}$

21. The force- time graph for the motion of a body is shown in the following figure. The change in momentum of the body between $0\ and\ 20\ s$ is



(A) $20\ kg\ ms^{-1}$ (B) $25\ kg\ ms^{-1}$

(C) $30\ kg\ ms^{-1}$ (D) $35\ kg\ ms^{-1}$

22. A vessel contains oil (density= $0.8\ g\ cm^{-3}$) over mercury (density= $13.6\ g\ cm^{-3}$). A homogeneous sphere floats with half its volume immersed in mercury and the other half in oil. The density of the material of the sphere in $g\ cm^{-3}$ is:

(A) 3.3 (B) 6.4

(C) 7.2 (D) 12.8

23. A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upwards with a net acceleration of $g/3$, the fraction of its volume immersed in the liquid is

(A) $1/2$ (B) $3/8$

(C) $2/3$ (D) $3/4$

Space for Rough work

24. The time period of a simple pendulum is T. The pendulum is oscillated with its bob immersed in a liquid of density σ. If the density of the bob is p and viscous effect is neglected, the time period of the pendulum will be

(A) $\left(\frac{p}{p-\sigma}\right)^{1/2} T$ (B) $\left(\frac{\sigma}{p-\sigma}\right)^{1/2} T$

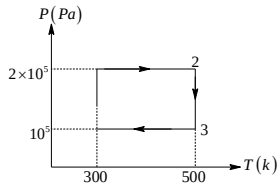
(C) $\left(\frac{p}{\sigma}\right)^{1/2} T$ (D) $\left(\frac{\sigma}{p}\right)^{1/2} T$

25. For hydrogen gas $C_p - C_v = x$ and for oxygen gas $C_p - C_v = y$ where C_p *and* C_v are molar specific heats. Then the relation between x *and* y is

(A) $x = 16\,y$ (B) $y = 16\,x$

(C) $x = 4\,y$ (D) $x = y$

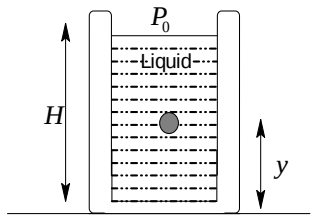
26. Two moles of helium gas are taken over the cycle 12341, as shown in the $p-T$ diagram. Assuming the gas to be ideal the work done on the gas in taking it from $1\,to\,2$ is



(A) $200\,R$ (B) $300\,R$

(C) $400\,R$ (D) $500\,R$

27. A small spherical monoatomic ideal gas bubble $\left[\gamma = \frac{5}{3}\right]$ is trapped inside a liquid of density p_L Assume that the bubble does not exchange any heat with the liquid. The bubble contains n moles of gas. The temperature of the gas when the bubble is at bottom is T_0. The height of the liquid is H and the atmospheric pressure is P_0. Neglect surface tension. As the bubble moves upwards, besides the buoyancy force the forces acting on it are besides the buoyancy force the forces acting on it are



(A) Only the force of gravity

(B) The force due to gravity and force due to pressure of the liquid

(C) The force due to gravity, the force due to pressure of the liquid and force due to viscosity of the liquid

(D) The force due to gravity and the force due to viscosity of the liquid

Space for Rough work

28. A knife edge of mass M is dropped from a height h on a wooden floor with its tip pointing of downward. If the blade penetrates a distance S into the wood the average resistance offered by the wood to the blade is

(A) Mg (B) $Mg\left(1+\frac{h}{s}\right)$

(C) $Mg\left(1-\frac{h}{s}\right)$ (D) $Mg\left(1+\frac{h}{s}\right)^2$

29. A uniform rope of mass M and length L is pulled along a frictionless inclined plane of angle(by appling a force $F\left(>Mg\right)$)parallel to incline. Tension in the chain at a distance x from the end at which force is applied is

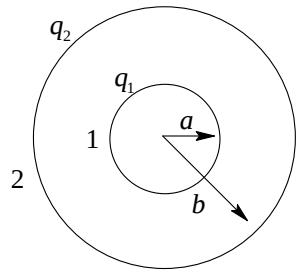
(A) $F\frac{x}{L}+\frac{x}{L}Mg\sin\theta$ (B) $F\left(1-\frac{x}{L}\right)$

(C) $F\left(1-\frac{x}{L}\right)+\frac{x}{L}Mg\sin\theta$ (D) None of these

30. Two Conducting spherical shells having radii a *and* b charged to q_1 & q_2 respectively. Find the potential difference between $1\&2$

(A) $\frac{q_1}{4\pi\in_0}\left(\frac{1}{a}-\frac{1}{b}\right)$ (B) $\frac{q_2}{4\pi\in_0}\left(\frac{1}{a}-\frac{1}{b}\right)$

(C) $\frac{q_1+q_2}{4\pi\in_0}\left(\frac{1}{a}-\frac{1}{b}\right)$ (D) $\frac{q_1}{4\pi\in_0 a}$



Space for Rough work

Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. $112\ mL$ of NO_2 at NTP was liquefied, the density of the liquid being $1.15\,g\ mL^{-1}$. The volume of the liquid is:
   (A) $0.1\ mL$ (B) $0.2\ mL$
   (C) $128.8\ mL$ (D) $1.15\ mL$

2. A mixture of $HCOOH$ and $H_2C_2O_4$ is heated with conc. H_2SO_4. The gaseous mixture thus produced is collected and passed through $NaOH$ solution. The volume of the gaseous mixture decreases by $\left(\frac{1}{6}\right)^{th}$. The molar ratio of $HCOOH$ and $H_2C_2O_4$ in the mixture of acids is:
   (A) $\frac{1}{4}$ (B) 6
   (C) 4 (D) $\frac{1}{6}$

3. Equal volumes of ethanoic acid and KOH are mixed, the heat evolved is:
   (A) equal to heat of formation of water from H^+ and OH^-
   (B) less than the heat of formation of water from H^+ and OH^-
   (C) equal to heat of formation of water from H_2 and O_2 in gaseous state
   (D) equal to enthalpy of neutralization of KOH and HCl

4. Consider the following reaction,
   $A(g) \longrightarrow 2B(g)$
   The rate constant for this reaction is $2.303\times10^{-2}\ \text{min}^{-1}$. The time in which the molar ratio of A to B becomes $3:2$ is equal to:
   (A) 12.5 min (B) 2.303 min
   (C) 5.12 min (D) 80 min

Space for Rough work

5. Consider the following gaseous phase equilibrium, $A(g) \rightleftharpoons 2B(g)$. The degree of dissociation of A is $\frac{1}{3}$ and $\frac{2}{3}$ at P_1 and P_2 pressures respectively. The ratio of P_1 and P_2 is:

(A) $\frac{7}{4}$ (B) $\frac{7}{32}$

(C) $\frac{8}{32}$ (D) $\frac{32}{5}$

6. The pH of a H_2SO_4 solution is 2. The pH of this solution, if equal volume of $0.2\,M$ $NaCl$ solution is added to it, will be:

(A) 1.7 (B) 2.3

(C) 3.2 (D) 2

7. In a crystal structure of an element, atoms are present in ccp arrangement and tetrahedral voids, which of the following relation between the atomic radius and edge length is correct for the unit cell which is formed in this arrangement? (a is the edge length and R is the radius of atom):

(A) $\sqrt{2}a = 4R$ (B) $\sqrt{3}a = 8R$

(C) $\sqrt{3}a = 4R$ (D) $a = 2R$

8. $1000\,g$ of 1 molal aqueous solution of sucrose is cooled and maintained at $-3.534°C$. What is the amount of ice separated out $\left(K_f \text{ for } H_2O \text{ is } 1.86\right)$:

(A) $353\,g$ (B) $220\,g$

(C) $150\,g$ (D) $510\,g$

9. Acidified water is electrolysed using inert electrodes. The total volume of gases collected at cathode and anode is $V\,mL$ at STP. The amount of charge passed in the electrolytic cell is equal to:

(A) $\left(5.74\times10^3 \times V\right)C$ (B) $96500\,C$

(C) $\left(1.72\times10^3 \times V\right)C$ (D) $1.72\times10^3 C$

Space for Rough work

10. The charge on As_2S_3 may be due to the adsorption of:
(A) H^+ (B) OH^-
(C) O_2^- (D) S^{2-}

11. The magnetic moment of Cr^{x+} is $\sqrt{15}\,BM$. The ratio of number of paired e^- and unpaired e^- in Cr^{x+} is equal to:
(A) seven (B) six
(C) five (D) four

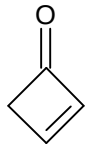
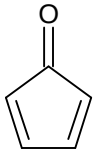
12. Find ΔH in terms of KJ for the following reaction from the given data
$M(g)+2X(g)\longrightarrow M^{2+}(g)+2X^-(g)$
$(IE)_1$ *of* $M(g)=705\,KJ\,mol^{-1}$
$(IE)_2$ *of* $M(g)=1151\,KJ\,mol^{-1}$
$(IA)_1$ *of* $X(g)=-328\,KJ\,mol^{-1}$
(A) $120\,KJ$ (B) $512\,KJ$
(C) $1200\,KJ$ (D) $110\,KJ$

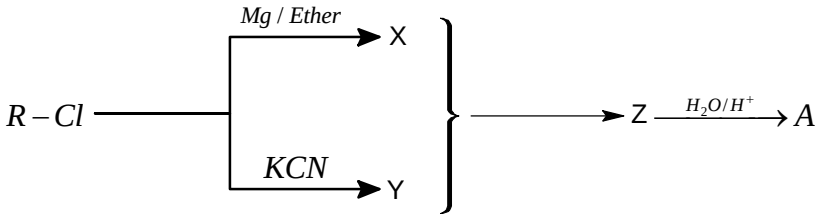
13. Which of the following is not three dimensional network structures?
(A) SiO_2 (B) CO_2
(C) diamond (D) graphite

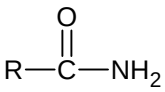
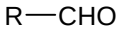
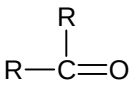
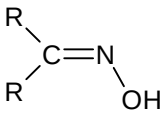
14. A 1g sample of H_2O_2 solution containing $x\%$ H_2O_2 by weight required $x\,mL$ of $KMnO_4$ for complete oxidation in acidic medium. The normality of $KMnO_4$ solution is approximately:
(A) 0.1 (B) 0.4
(C) 0.6 (D) 1.2

Space for Rough work

15. Consider the following statements about the structure of borax:
(i) number of $B-O-B$ bonds is x
(ii) number of $B-B$ bonds is y
(iii) number of sp^2 hybridized B atoms is z
The value of $(x+y+z)$ is equal to
(A) 3 (B) 4
(C) 6 (D) 7

16. Which of the following has maximum dipole moment?
(A) 
(B) 
(C) 
(D) All have equal value

17. 

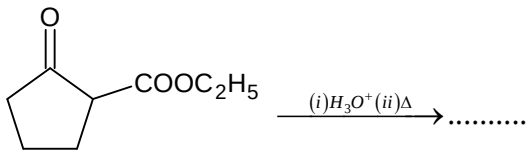
What is 'A'?
(A) 
(B) 
(C) 
(D) 

Space for Rough work

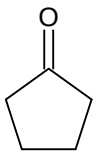
18. Which of the following will not convert $R-C\equiv N$ to $R-CHO$?

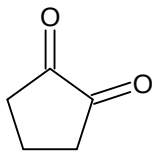
(A) $(i)\,SnCl_2 / HCl$ (B) $(i)\,RMgX \quad (ii)\,H_2O / H^+$

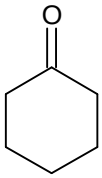
(C) $(i)\,LiAlH_4 \quad (ii)\,H_2O / \Delta$ (D) $(i)\,AlH(i-Bu)_2 \,(ii)\,H_2O / \Delta$

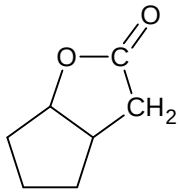
19. 

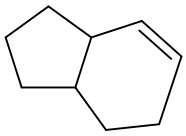
The major product is:

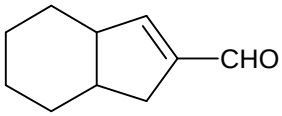
(A) 

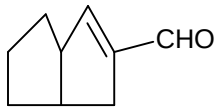
(B) 

(C) 

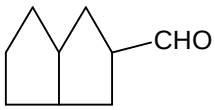
(D) 

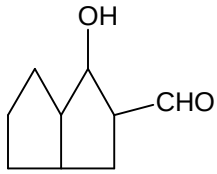
20. 
$\xrightarrow[Zn/H_2O]{O_3} X \xrightarrow{KOH/\Delta} Y$; the compound '$Y$' is :

(A) 

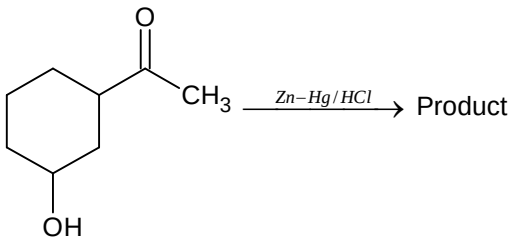
(B) 

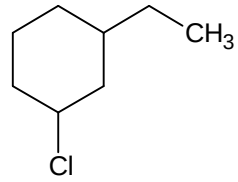
x

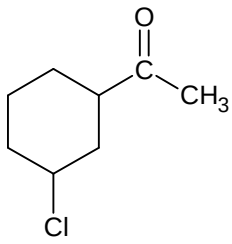
(C) 

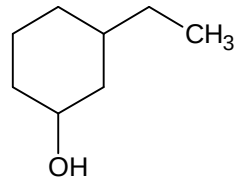
(D) 

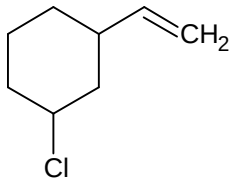
Space for Rough work

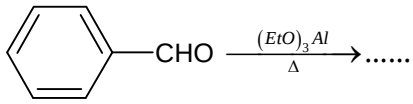
21. 

(A) 

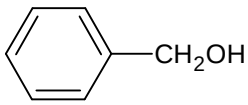
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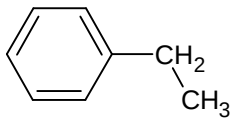
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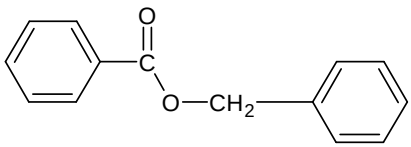
(D) 

22. 

The major product is:

(A) 

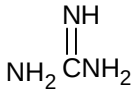
(B) 

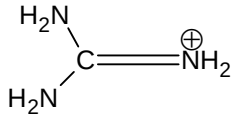
(C) 

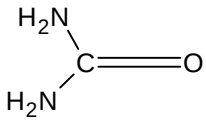
(D) No reaction

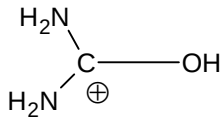
Space for Rough work

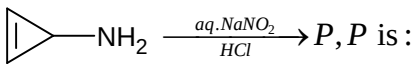
23. Strongest base is:

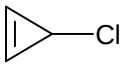
(A) 

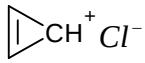
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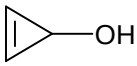
(C) 

(D) 

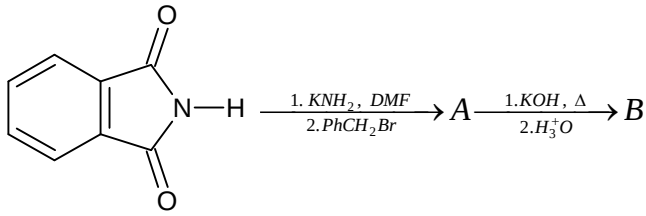
24. 

(A) 

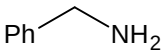
(B) 

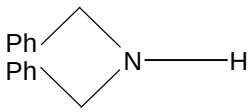
(C) 

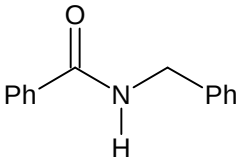
(D) None of these

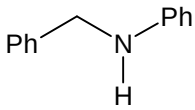
25. 

The end product B of the above reaction is:

(A) 

(B) 

(C) 

(D) 

Space for Rough work

26. Marshall's acid is:
(A) $H_2S_2O_5$ (B) $H_2S_2O_8$
(C) H_2SO_4 (D) $H_2S_2O_7$

27. The correct order of $S-O$ bond in $(I)\,SO_2$, $(II)\,SO_3$, $(III)\,SO_4^{2-}$ is:
(A) $I > II > III$ (B) $III > I > II$
(C) $II > III > I$ (D) $I > III > II$

28. The following acids have been arranged in the order of increasing acid strength. Identify the correct order:
$(I)\,ClOH$ $(II)\,BrOH$
$(III)\,IOH$
(A) $I < II < III$ (B) $III < II < I$
(C) $II < I < III$ (D) $III < I < II$

29. The correct order of oxidizing power is:
(A) $HOCl > HOClO > HOClO_2 > HOClO_3$ (B) $HOClO_3 > HOClO_2 > HOClO > HOCl$
(C) $HOClO_2 > HOClO_3 > HOCl > HOClO$ (D) $HOCl > HOClO_2 > HOClO > HOClO_3$

30. Shape of $XeOF_4$ is:
(A) Trigonal bipyramid with lp at axial position
(B) Square pyramid with lp at axial position
(C) Distorted square pyramid with lp at axial position
(D) Square pyramid with lp at square plane.

Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. The number of ordered pairs (x, y) of real numbers satisfying $4x^2 - 4x + 2 = \sin^2 y$ and $x^2 + y^2 \le 3$ is equal to
   (A) 0 (B) 2
   (C) 4 (D) 8

2. $\sin 47° - \sin 25° + \sin 61° - \sin 11° =$
   (A) $\cos 7°$ (B) $\sin 7°$
   (C) $2\cos 7°$ (D) $2\sin 7°$

3. The tangent to the hyperbola $xy = c^2$ at the point P intersects the x-axis at T and the y-axis at T'. The normal to the hyperbola at P intersects the x-axis at N and the y-axis at N'. The areas of the triangles PNT and PN'T' are Δ and Δ' respectively , then $\frac{1}{\Delta} + \frac{1}{\Delta'}$ is
   (A) equal to 1 (B) depends on t
   (C) depends on c (D) equal to 2

4. The normal at P to a hyperbola of eccentricity e, intersects its transverse and conjugate axes at L and M respectively. If locus of the mid-point of LM is hyperbola, then eccentricity of the hyperbola is:
   (A) $\left(\frac{e+1}{e-1}\right)$ (B) $\frac{e}{\sqrt{(e^2-1)}}$
   (C) e (D) None of these

Space for Rough work

5. Area of the triangle formed by the lines $x-y=0, x+y=0$ and any tangent to the hyperbola $x^2-y^2=a^2$ is:

(A) $|a|$ (B) $\frac{1}{2}|a|$

(C) a^2 (D) $\frac{1}{2}a^2$

6. A rectangular hyperbola whose centre is C is cut by any circle of radius r in four points P, Q, R and S. Then $CP^2+CQ^2+CR^2+CS^2$ is equal to:

(A) r^2 (B) $2r^2$

(C) $3r^2$ (D) $4r^2$

7. The point, at shortest distance from the line $x+y=7$ and lying on an ellipse $x^2+2y^2=6$, has co-ordinates:

(A) $\left(\sqrt{2},\sqrt{2}\right)$ (B) $\left(0,\sqrt{3}\right)$

(C) $(2,1)$ (D) $\left(\sqrt{5},\frac{1}{\sqrt{2}}\right)$

8. Locus of the foot of perpendicular drawn from origin to any tangent to the ellipse $\frac{x^2}{2}+y^2=1$ is:

(A) $\left(x^2+y^2\right)^2=2x^2+y^2$ (B) $\left(x^2-y^2\right)^2=2x^2+y^2$

(C) $\left(x^2+y^2\right)^2=2x^2-y^2$ (D) None of these

9. The line $5x-3y=8\sqrt{2}$ is a normal to the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$. If θ be the eccentric angle of the foot of this normal, then 'θ' is equal to:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$ (D) None of these

Space for Rough work

10. For parabola $x^2 + y^2 + 2xy - 6x - 2y + 3 = 0$, then focus is:

(A) $(1,-1)$ (B) $(-1,1)$

(C) $(3,1)$ (D) None of these

11. The triangle formed by the tangent to the parabola $y = x^2$ at the point whose abscissa is x_0 $(x_0 \in [1,2])$, the y-axis and the straight line $y = x_0^2$ has the greatest area if x_0 is equal to:

(A) 0 (B) 1

(C) 2 (D) 3

12. The condition that the parabola $y^2 = 4c(x-d)$ *and* $y^2 = 4ax$ have a common normal other than x- axis $(a > 0, c > 0)$ is:

(A) $2a < 2c + d$ (B) $2c < 2a + d$

(C) $2d < 2a + c$ (D) $2d < 2c + a$

13. The point $([P+1],[P])$, (where $[.]$ denotes the greatest integer function) lying inside the region bounded by the circle $x^2 + y^2 - 2x - 15 = 0$ *and* $x^2 + y^2 - 2x - 7 = 0$ then:

(A) $p \in [-1,0) \cup [0,1) \cup [1,2)$ (B) $p \in [-1,2) - \{0,1\}$

(C) $p \in (-1,2)$ (D) None of the above

14. If the distance from the origin of the centres of three, circles $x^2 + y^2 + 2\lambda_i x - c^2 = 0, (i = 1,2,3)$ are in GP. then the lengths of the tangents drawn to them from any point on the circle $x^2 + y^2 = c^2$ are in:

(A) A. P (B) G. P

(C) H.P (D) None of these

Space for Rough work

15. The values of k for which the circle $x^2+y^2=1 and\ x^2+y^2-4\lambda x+8=0$ have two common tangents is:

(A) $\left(-\frac{9}{4},\frac{9}{4}\right)$ (B) $\left(-\infty,-\frac{9}{4}\right]\cup\left[\frac{9}{4},\infty\right)$

(C) $\left(-\infty,-\frac{9}{4}\right)\cup\left(\frac{9}{4},\infty\right)$ (D) None of these

16. The equation of image of pair of lines $y=|x-1|$ in y- axis is:

(A) $x^2+y^2+2x+1=0$ (B) $x^2-y^2+2x-1=0$

(C) $x^2-y^2+2x+1=0$ (D) None of these

17. If $\alpha,\beta>0 and\ \alpha<\beta and\ ax^2+4\gamma xy+\beta y^2+4p(x+y+1)=0$ represent a pair of straight lines, then:

(A) $\alpha\le p\le\beta$ (B) $p\le\alpha$

(C) $p\ge\alpha$ (D) $p\le\alpha\ or\ p\ge\beta$

18. If $\frac{2}{1!9!}+\frac{2}{3!7!}+\frac{4}{5!5!}=\frac{2^m}{n!}$, then orthocentre of the triangle having sides $x-y+1=0, x+y+3=0 and\ 2x+5y-2=0$ is:

(A) $(2m-2n, m-n)$ (B) $(2m-2n, n-m)$

(C) $(2m-n, m+n)$ (D) $(2m-n, m-n)$

19. In ΔABC, if $p=\tan\frac{B-C}{2}\cot\frac{B+C}{2}$ $q=\tan\frac{C-A}{2}\cot\frac{C+A}{2}$ and $r=\tan\frac{A-B}{2}\cot\frac{A+B}{2}$ then $\sum p=$:

(A) 1 (B) $\sum pq$

(C) $-pqr$ (D) None of these

Space for Rough work

20. In the triangle ABC the medians from B and C are perpendicular. The value of $\cot B + \cot C$ cannot be:

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{4}{3}$ (D) $\frac{5}{3}$

21. If $ABCD$ be a cyclic quadrilateral inscribed in a circle of radius 2 unit such that $BD = 2\sqrt{3}$; $AB = 1$ then $AD =$:

(A) $\frac{1+3\sqrt{5}}{2}$ (B) $\frac{3\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{5}+1}{2}$ (D) None of these

22. If I be the incentre of ΔABC and P, Q are feet of perpendicular from A to BI, CI respectively. Then $AQ : CI + AP : BI =$:

(A) $\cot\left(\frac{A+B}{2}\right)$ (B) $\cot\left(\frac{B+C}{2}\right)$

(C) $\cot\left(\frac{C+A}{2}\right)$ (D) $\tan\left(\frac{B+C}{2}\right)$

23. Number of roots of the equation $\cos^2 x + \left(\frac{\sqrt{3}+1}{2}\right)\sin x - \frac{\sqrt{3}}{4} - 1 = 0$ which lie in the interval $[-\pi, \pi]$ is:

(A) 2 (B) 4

(C) 6 (D) 8

24. The number of solutions of $\tan(5\pi\cos\theta) = \cot(5\pi\sin\theta)$ for $\theta \in (0, 2\pi)$ is:

(A) 28 (B) 14

(C) 4 (D) 2

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25. Number of solutions of the equation $4\sin^2 x+\tan^2 x+\cot^2 x+\cos ec^2 x=6$ in $[0,\pi]$ is:
(A) 0 (B) 2
(C) 8 (D) 4

26. The equation $2x=(2n+1)\pi(1-\cos x)$ has (where n is a positive integer):
(A) infinity many real roots (B) exactly one real root
(C) exactly $2n+2$ real roots (D) exactly $2n+3$ real roots

27. A vertical tower standing at O has marks P, Q, R, S at heights of 1m, 2m, 3m, and 4m from the foot O and A is a point on the horizontal plane through O. If PQ and RS subtend angles α and β respectively at A where OA = 2 m then $\cos(\alpha+\beta)=$:
(A) $5/\sqrt{26}$ (B) $24/\sqrt{650}$
(C) $23/\sqrt{650}$ (D) $1/\sqrt{26}$

28. A,B,C are three points on a horizontal line through the base O of a pillar OP, such that OA, OB, OC are in A. P. If α,β,γ the angles of elevation of the top of the pillar at A,B,C respectively are also in A. P. then $\sin\alpha,\sin\beta,\sin\gamma$ are in:
(A) A. P. (B) G. P.
(C) H. P. (D) None of these

29. If $n\in N$, then $7^{2n}+2^{3n-3}.3^{n-1}$ is always divisible by:
(A) 25 (B) 35
(C) 45 (D) None of these

30. Let $P(n)$ denotes the statement that n^2+n is odd. It is seen that $P(n)\Rightarrow P(n+1).P_n$ is true for all:
(A) $n>1$ (B) $n\in N$
(C) $n>2$ (D) None of these

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